

PALMAS, Brazil

WMO: 830650

Lat: 10.283S

Lon: 48.350W

Elev: 774

StdP: 14.29

Time Zone: -3.00 (W03)

Period: 06-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	66.2	67.9	44.9	45.2	90.3	47.3	49.6	87.6	16.0	79.4	13.7	80.3	4.6	100	0.318

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
9	21.6	98.7	69.7	97.1	69.7	96.4	69.8	78.9	86.6	78.3	86.1	77.8	85.7	7.9	300

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
77.1	145.1	81.7	75.8	139.0	81.0	75.5	137.3	80.8	42.9	87.1	42.3	86.3	41.7	85.5	81.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
15.6	13.0	11.4	DB	62.9	101.7	1.8	2.3	61.7	103.4	60.6	104.7	59.6	106.0	58.3	107.6
			WB	56.6	79.9	1.6	1.1	55.4	80.7	54.4	81.3	53.5	81.9	52.3	82.7

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	81.7	80.2	80.1	80.7	81.3	81.6	80.6	80.6	83.3	85.9	83.9	81.8	80.9
	DBStd	2.89	2.37	2.72	2.34	1.92	1.83	1.60	1.79	2.22	2.37	3.28	2.71	2.61
	HDD50	0	0	0	0	0	0	0	0	0	0	0	0	0
	HDD65	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDD50	11588	936	844	951	940	979	919	948	1031	1077	1051	953	959
	CDD65	6113	471	424	486	490	514	469	483	566	627	586	503	494
	CDH74	62607	3837	3445	4187	4435	5147	4918	5398	7041	8263	6657	4859	4420
	CDH80	28809	1322	1211	1450	1697	2346	2480	2865	3943	4655	3263	1911	1666
Wind	WSAvg	5.0	4.5	4.5	4.2	4.3	4.8	5.6	6.0	6.1	5.8	5.1	4.8	4.6
Precipitation	PrecAvg	67.7	11.9	10.7	10.6	6.3	2.2	0.1	0.0	0.2	1.9	5.0	8.2	10.6
	PrecMax	82.9	23.6	15.6	19.8	11.8	7.1	1.0	0.3	1.1	4.8	10.0	12.4	20.2
	PrecMin	50.0	5.6	3.0	5.9	2.1	0.2	0.0	0.0	0.2	2.1	4.0	4.8	4.8
	PrecStd	9.0	5.0	3.2	3.5	2.3	1.4	0.2	0.1	0.2	1.4	2.0	2.1	4.0
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	93.6	96.4	93.4	94.6	93.6	94.8	95.4	98.8	100.7	100.5	97.0	95.4
		MCWB	74.6	74.7	76.7	74.2	73.6	72.0	68.5	68.5	68.5	69.9	73.8	74.5
	2%	DB	91.5	92.8	91.3	92.8	93.1	93.2	94.7	97.1	98.9	98.5	93.6	93.1
		MCWB	75.7	75.5	76.4	75.0	73.5	70.7	68.5	68.2	69.1	70.9	74.8	75.3
	5%	DB	89.4	89.5	89.5	91.0	91.6	91.7	93.3	95.4	98.2	96.6	91.5	90.1
		MCWB	75.5	75.6	76.2	75.8	73.8	70.3	68.3	68.3	69.2	71.2	75.2	75.7
	10%	DB	86.8	86.4	87.6	88.2	90.0	91.2	91.7	94.7	96.6	93.5	89.3	88.0
		MCWB	75.7	75.7	76.3	75.9	73.8	70.3	68.3	68.2	69.1	72.5	75.3	75.7
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	78.9	78.9	79.7	79.1	78.6	76.9	73.9	74.3	76.5	78.1	79.0	79.1
		MCDB	86.5	86.7	86.7	86.1	86.7	86.1	86.4	88.2	88.2	87.8	87.4	87.8
	2%	WB	78.0	77.9	78.7	78.3	77.5	75.4	72.3	72.7	75.5	77.0	77.8	78.1
		MCDB	85.6	85.7	85.9	85.5	86.3	86.0	87.1	87.9	87.1	87.5	86.7	86.8
	5%	WB	77.1	77.1	78.0	77.8	76.6	74.1	70.9	71.2	74.7	76.1	77.0	77.2
		MCDB	84.8	84.9	85.1	85.0	85.7	85.4	87.1	88.0	86.6	86.2	86.3	85.7
	10%	WB	76.3	76.3	77.3	77.1	75.9	73.0	69.8	70.0	73.7	75.4	76.2	76.4
		MCDB	83.8	83.8	84.3	84.3	85.0	84.7	86.8	88.9	86.3	85.1	85.1	84.5
Mean Daily Temperature Range	5% DB	MDBR	14.6	14.2	14.2	15.4	18.1	22.1	24.5	24.2	21.6	18.1	15.3	15.0
		MCDBR	18.0	17.9	17.1	18.6	20.1	23.2	25.7	25.2	24.1	22.6	18.9	18.3
		MCWBR	6.2	6.4	6.2	6.5	6.4	7.5	8.1	8.7	8.9	7.3	6.2	6.3
	5% WB	MCDBR	15.5	15.5	15.2	15.7	17.5	21.2	24.9	24.5	20.6	17.8	16.5	16.5
		MCWBR	6.4	6.4	6.2	6.1	6.4	8.0	9.0	9.2	7.5	6.1	6.1	6.5
Clear-Sky Solar Irradiance	taub		0.406	0.405	0.404	0.401	0.389	0.353	0.351	0.402	0.517	0.521	0.459	0.437
	taud		2.456	2.465	2.476	2.464	2.491	2.574	2.550	2.365	2.036	2.083	2.281	2.352
	Ebn at Noon		297	296	291	282	276	282	285	278	255	260	279	287
	Edh at Noon		38	38	36	35	32	29	30	38	56	55	45	42
All-Sky Solar Radiation	RadAvg		1662	1645	1605	1649	1664	1735	1838	1981	1915	1753	1660	1660
	RadStd		156	190	79	107	89	50	48	84	151	133	116	110

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (0 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page